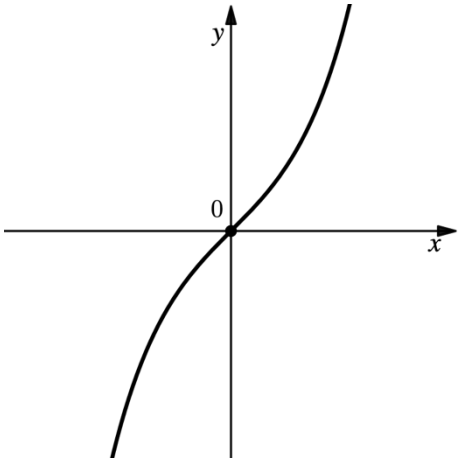
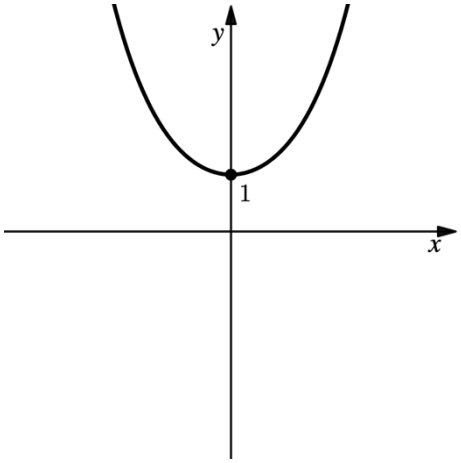
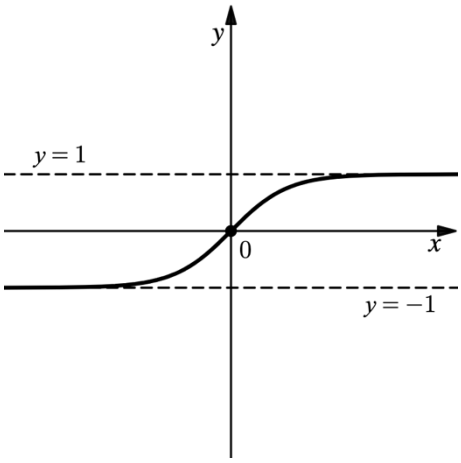
1) a)



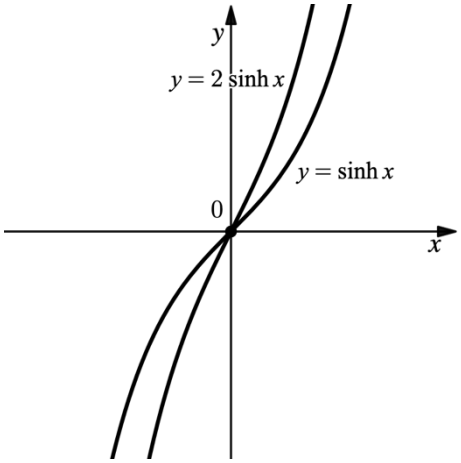
1) b)



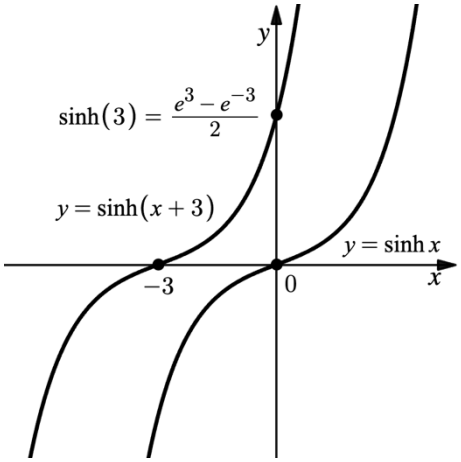
1) c)



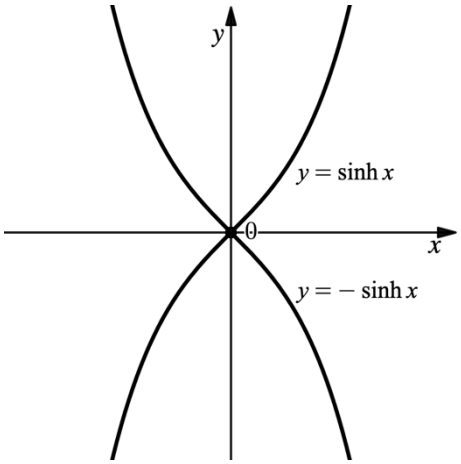
2) a)


y
y = 2 sinh x
y = sinh x
0
x

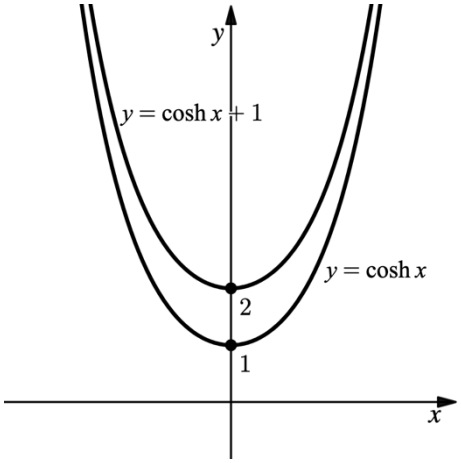
2) b)


y
sinh(3) = (e^3 − e^−3)/2
y = sinh(x + 3)
y = sinh x
−3
0
x

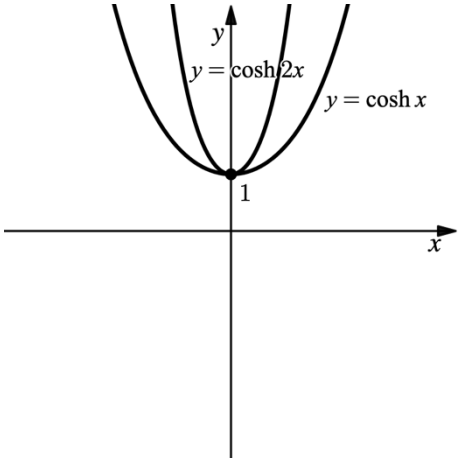
2) c)


y
y = sinh x
0
x
y = − sinh x

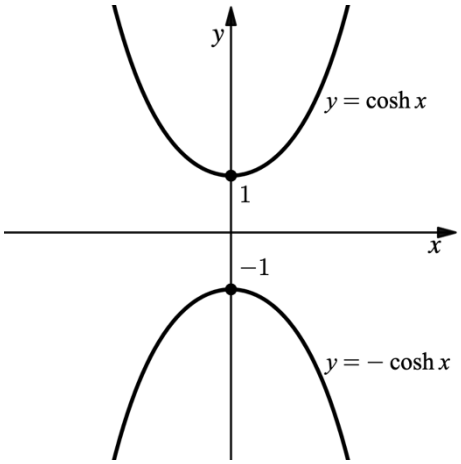
2) d)



2) e)



2) f)



3) a)

$y = \sinh x + b$

Substitute $\left(\ln\left(\sqrt{2}-1\right), 0\right)$:

$0 = \sinh\left(\ln\left(\sqrt{2}-1\right)\right) + b$

$\Rightarrow 0 = \frac{e^{\ln(\sqrt{2}-1)} - e^{-\ln(\sqrt{2}-1)}}{2} + b$

$\Rightarrow 0 = e^{\ln(\sqrt{2}-1)} - e^{\ln(\sqrt{2}-1)^{-1}} + 2b$

$\Rightarrow 0 = \sqrt{2} - 1 - \frac{1}{\sqrt{2}-1} + 2b$

$\Rightarrow 0 = \sqrt{2} - 1 - \left(1 + \sqrt{2}\right) + 2b$

$\Rightarrow 0 = \sqrt{2} - 1 - 1 - \sqrt{2} + 2b$

$\Rightarrow 2b = 2$

$\Rightarrow b = 1$

$y = \sinh x + 1$

3) b)

$y = \sinh(x - a)$

Substitute $\left(0, \frac{1-e^4}{2e^2}\right)$

$\frac{1-e^4}{2e^2} = \sinh(-a)$

$\Rightarrow \frac{1-e^4}{2e^2} = \frac{e^{-a} - e^{-(-a)}}{2}$

$\Rightarrow \frac{1-e^4}{e^2} = e^{-a} - e^{a}$

$\Rightarrow \frac{1}{e^2} - \frac{e^4}{e^2} = e^{-a} - e^{a}$

$\Rightarrow e^{-2} - e^{2} = e^{-a} - e^{a}$

$\Rightarrow a = 2$

$y = \sinh(x - 2)$

4) a)

$y = \cosh x + b$

Substitute $\left(\ln\left(3+2\sqrt{2}\right), 0\right)$:

$0 = \cosh\left(\ln\left(3+2\sqrt{2}\right)\right) + b$

$0 = \frac{e^{\ln(3+2\sqrt{2})} + e^{-\ln(3+2\sqrt{2})}}{2} + b$

$\Rightarrow 0 = e^{\ln(3+2\sqrt{2})} + e^{-\ln(3+2\sqrt{2})} + 2b$

$\Rightarrow 0 = e^{\ln(3+2\sqrt{2})} + e^{\ln(3+2\sqrt{2})^{-1}} + 2b$

$\Rightarrow 0 = 3 + 2\sqrt{2} + \frac{1}{3+2\sqrt{2}} + 2b$

$\Rightarrow 0 = 3 + 2\sqrt{2} + 3 - 2\sqrt{2} + 2b$

$\Rightarrow 0 = 6 + 2b$

$\Rightarrow b = -3$

$\therefore y = \cosh x - 3$

4) b)

$y = \cosh(x - a)$

Substitute $\left(0, \frac{5}{4}\right)$:

$\frac{5}{4} = \cosh(-a)$

$\Rightarrow \frac{5}{4} = \frac{e^{-a} + e^{-(-a)}}{2}$

$\Rightarrow \frac{5}{2} = e^{-a} + e^{a}$

$\Rightarrow \frac{5}{2}e^{a} = 1 + e^{2a}$

$\Rightarrow 0 = e^{2a} - \frac{5}{2}e^{a} + 1$

$\Rightarrow 0 = 2e^{2a} - 5e^{a} + 2$

$\Rightarrow 0 = (2e^{a} - 1)(e^{a} - 2)$

$\Rightarrow e^{a} = \frac{1}{2}$ or $e^{a} = 2$

$a = \ln\frac{1}{2}$ or $a = \ln 2$

$a = \pm \ln 2$

$\therefore y = \cosh(x + \ln 2)$

5) a)

$y = \tanh x$ has horizontal asymptotes at $y = -1$ and $y = 1$

$y = \tanh x \mapsto y = \tanh(x - 2) + 5$ is a translation by the vector $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$, so the horizontal asymptotes are translated 5 units upwards.

$\therefore y = \tanh(x - 2) + 5$ has horizontal asymptotes at $y = 4$ and $y = 6$

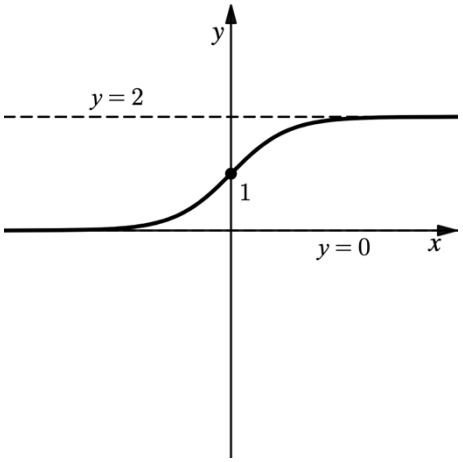
5) b)

$y = \tanh x$ has horizontal asymptotes at $y = -1$ and $y = 1$

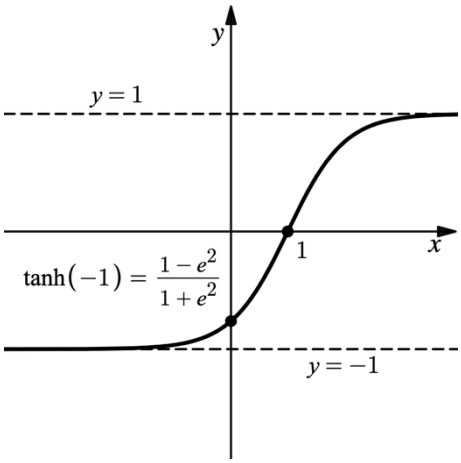
$y = \tanh x \mapsto y = \tanh\left(x + \frac{5}{2}\right) - \frac{3}{2}$ is a translation by the vector $\begin{pmatrix} -\frac{5}{2} \\ -\frac{3}{2} \end{pmatrix}$, so the horizontal asymptotes are translated $\frac{3}{2}$ units downwards.

$\therefore y = \tanh\left(x + \frac{5}{2}\right) - \frac{3}{2}$ has horizontal asymptotes at $y = -\frac{5}{2}$ and $y = -\frac{1}{2}$

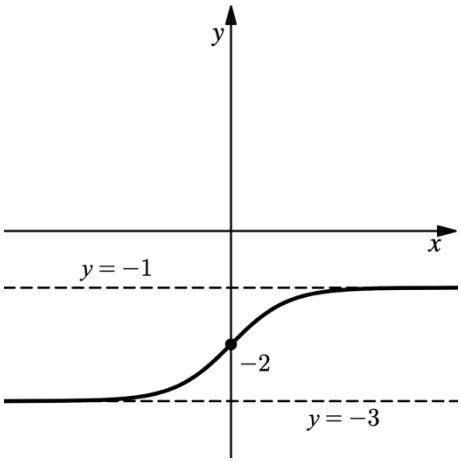
6) a)



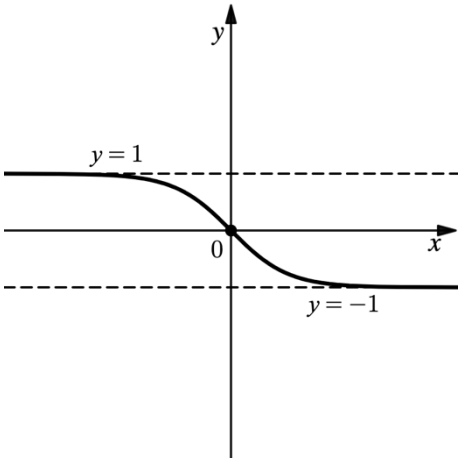
6) b)



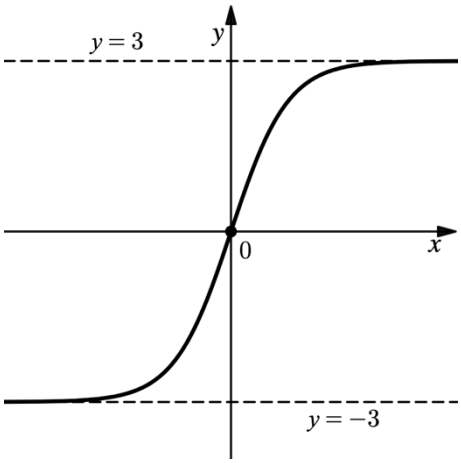
6) c)



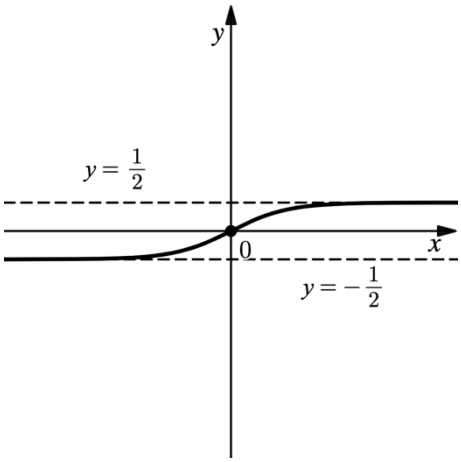
6) d)



7) a)



7) b)



8) a)

$\frac{3+1}{2} = 2$

$y = \tanh x + 2$

8) b)

$\frac{\frac{1}{2}+\left(-\frac{3}{2}\right)}{2} = -\frac{1}{2}$

$y = \tanh x - \frac{1}{2}$

8) c)

$y = \tanh(x - a)$

Substitute $\left(0, \frac{e^4-1}{e^4+1}\right)$:

$\frac{e^4-1}{e^4+1} = \tanh(-a)$

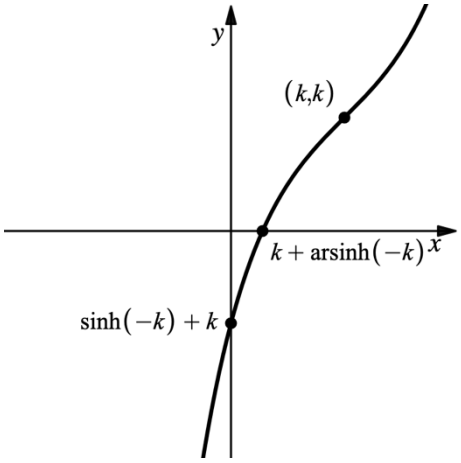
$\Rightarrow \frac{e^4-1}{e^4+1} = \frac{e^{-2a}-1}{e^{-2a}+1}$

$\Rightarrow -2a = 4$

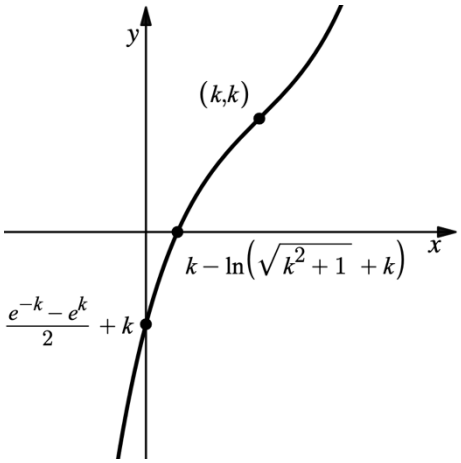
$\Rightarrow a = -2$

$\therefore y = \tanh(x + 2)$

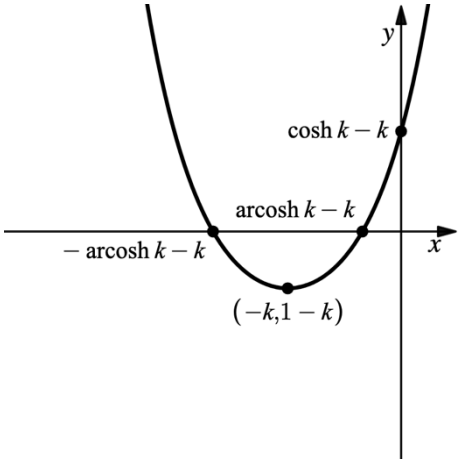
9) a)



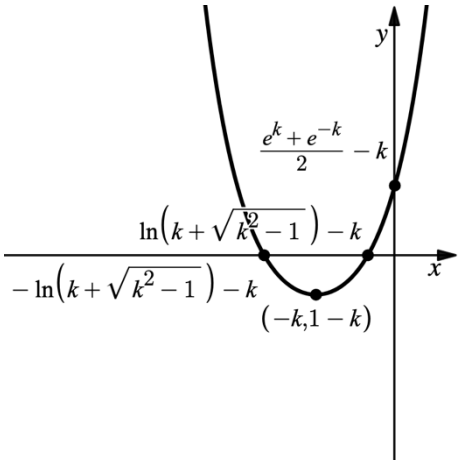
Or



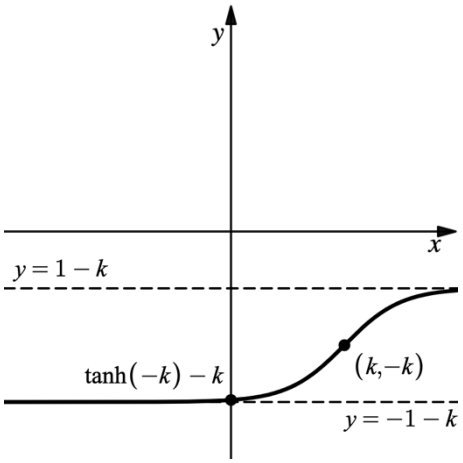
9) b)


y
cosh k − k
arcosh k − k
− arcosh k − k
x
(−k,1 − k)

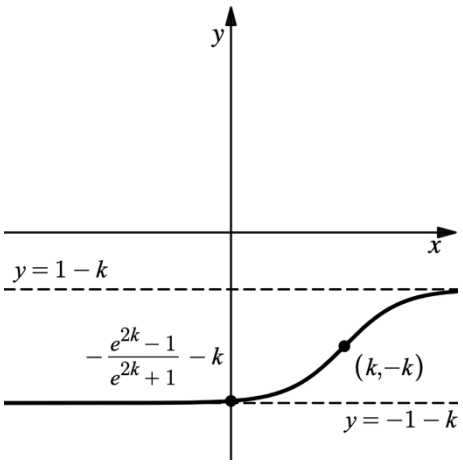
Or


y
$\frac{e^k + e^{-k}}{2} - k$
$\ln\left(k + \sqrt{k^2 - 1}\right) - k$
x
$-\ln\left(k + \sqrt{k^2 - 1}\right) - k$
(−k,1 − k)

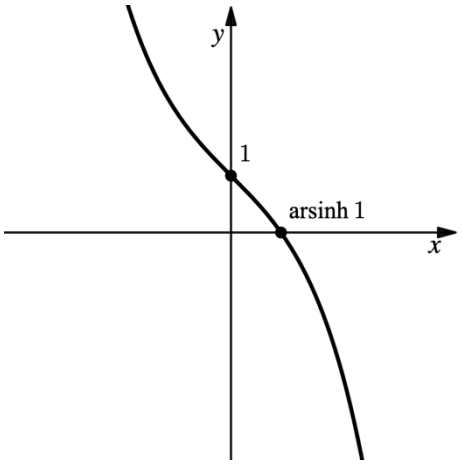
9) c)



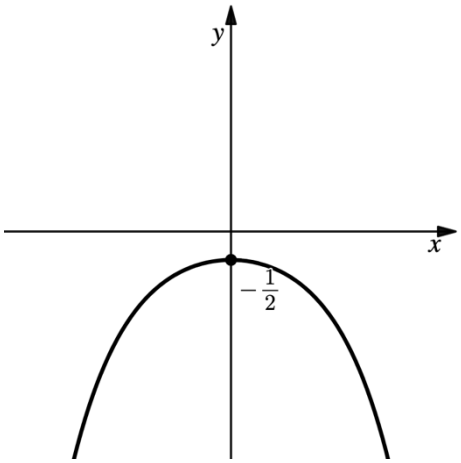
Or



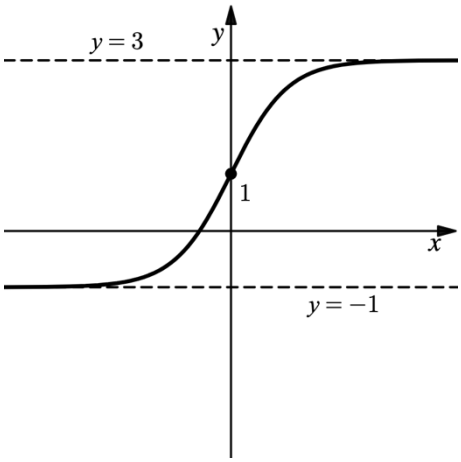
10) a)



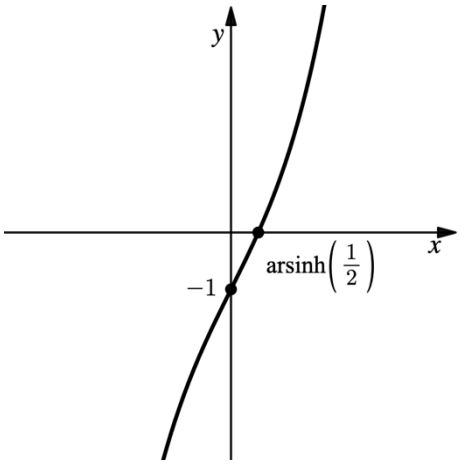
10) b)



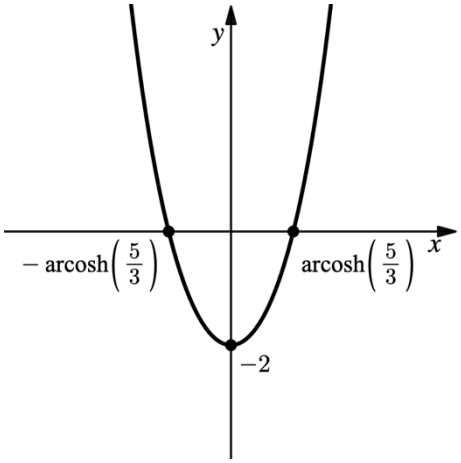
10) c)



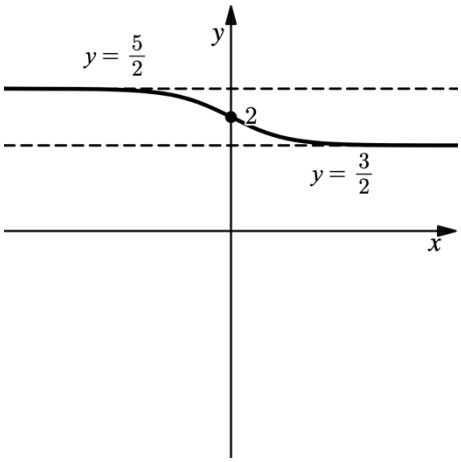
10) d)



10) e)

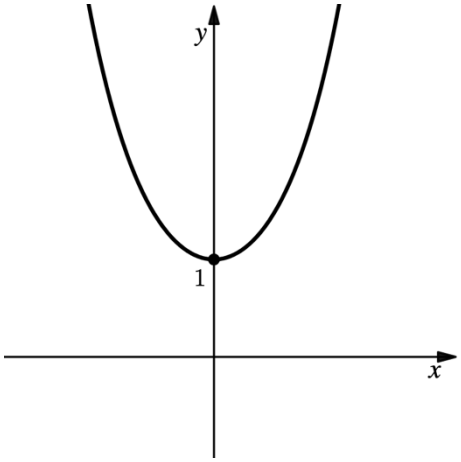

y
x
$-\operatorname{arcosh}\left(\frac{5}{3}\right)$
$\operatorname{arcosh}\left(\frac{5}{3}\right)$
-2

10) f)

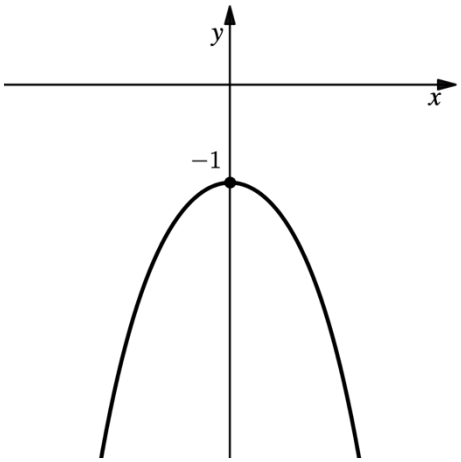

y
$y=\frac{5}{2}$
2
$y=\frac{3}{2}$
x

11) a)

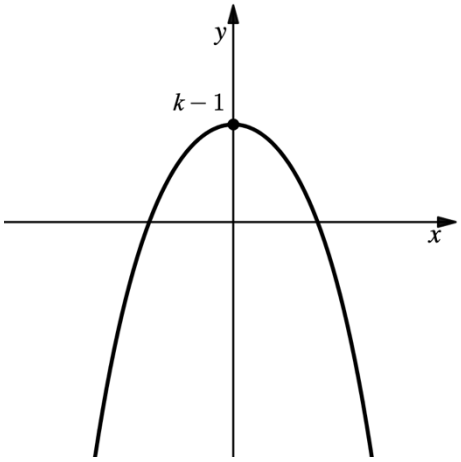
First consider $y = \cosh x$:



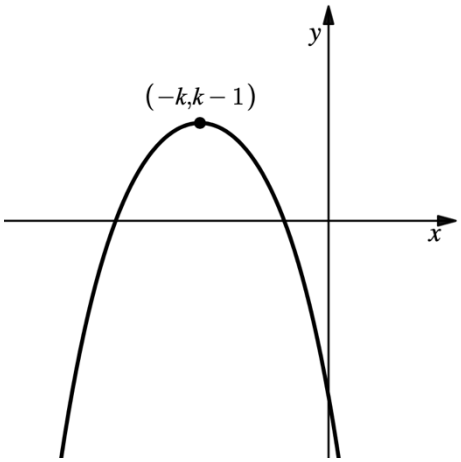
Reflect in the x-axis to get $y = -\cosh x$:



Translate by $\begin{pmatrix} 0 \\ k \end{pmatrix}$ to get $y = k - \cosh x$:



Translate by $\begin{pmatrix} -k \\ 0 \end{pmatrix}$ to get $y = k - \cosh(x + k)$:



Check the curve pass through $\left(0, k - \frac{e^k + e^{-k}}{2}\right)$:

$k - \cosh(0 + k) = k - \cosh k = k - \frac{e^k + e^{-k}}{2}$ as required

Alternative Method:

Using the y-intercept: $k - \frac{e^k + e^{-k}}{2} \equiv k - \cosh k$

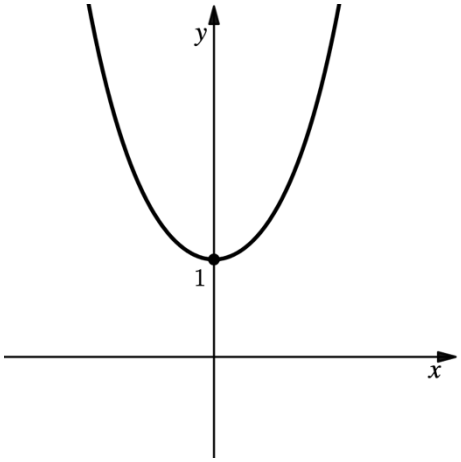
This must be the value when $x = 0$, hence: $y = k - \cosh(x + k)$

Check the curve pass through $(-k, k - 1)$:

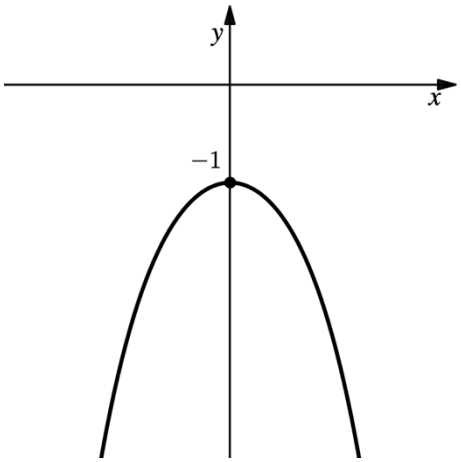
$k - \cosh(-k + k) = k - \cosh 0 = k - 1$ as required

11) b)

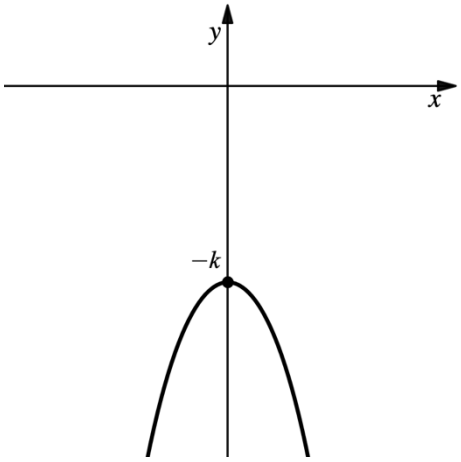
First consider $y = \cosh x$:



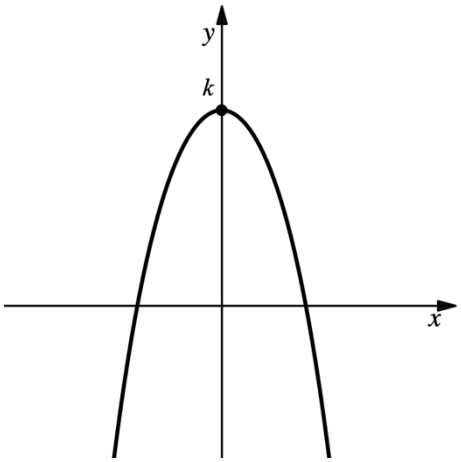
Reflect in the x-axis to get $y = -\cosh x$:



Stretch, parallel to the y-axis, scale factor k to get $y = -k\cosh x$:



Translate by $\begin{pmatrix} 0 \\ 2k \end{pmatrix}$ to get $y = 2k - k\cosh x$:



Check the curve passes through $\left(\ln\left(2+\sqrt{3}\right),0\right)$:

$2k-k\cosh\left(\ln\left(2+\sqrt{3}\right)\right)$

$=2k-k\times\frac{e^{\ln\left(2+\sqrt{3}\right)}+e^{-\ln\left(2+\sqrt{3}\right)}}{2}$

$=2k-k\times\frac{e^{\ln\left(2+\sqrt{3}\right)}+e^{\ln\left(2+\sqrt{3}\right)^{-1}}}{2}$

$=2k-\frac{1}{2}k\left(e^{\ln\left(2+\sqrt{3}\right)}+e^{\ln\left(2+\sqrt{3}\right)^{-1}}\right)$

$=2k-\frac{1}{2}k\left(2+\sqrt{3}+\frac{1}{2+\sqrt{3}}\right)$

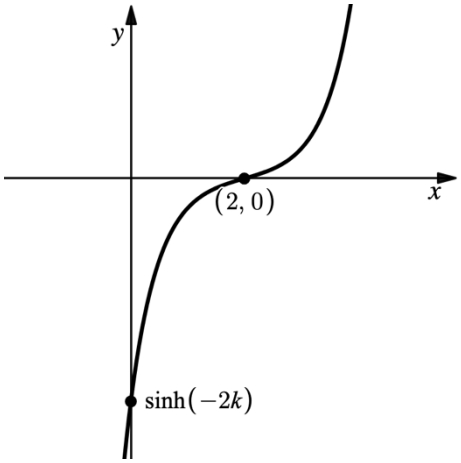
$=2k-\frac{1}{2}k\left(2+\sqrt{3}+\frac{1}{2+\sqrt{3}}\times\frac{2-\sqrt{3}}{2-\sqrt{3}}\right)$

$=2k-\frac{1}{2}k\left(2+\sqrt{3}+\frac{2-\sqrt{3}}{4-3}\right)$

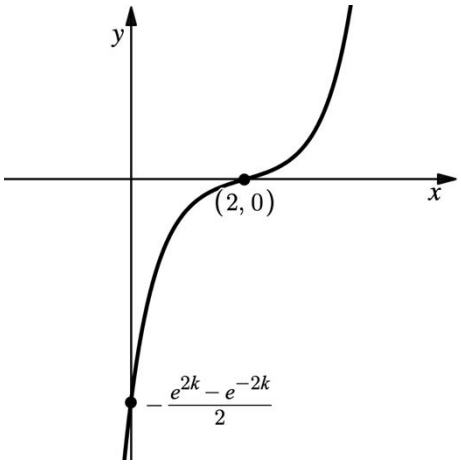
$=2k-\frac{1}{2}k\left(2+\sqrt{3}+2-\sqrt{3}\right)$

$=2k-\frac{1}{2}k\times4=0$ as required

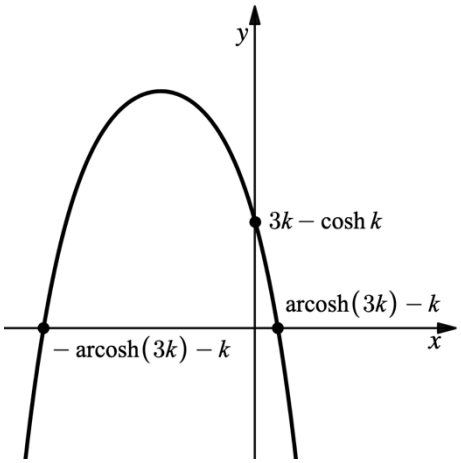
12) a)



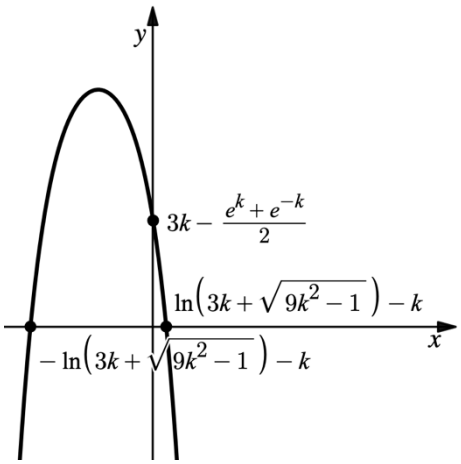
Or



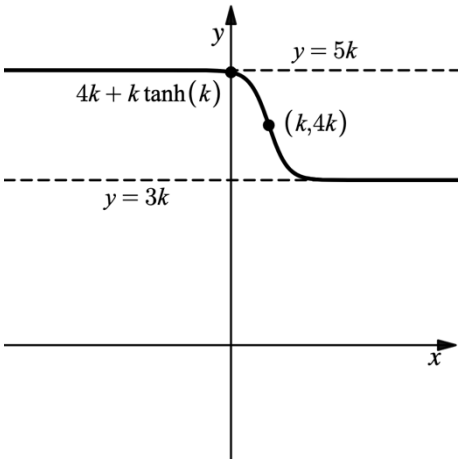
12) b)


y
$3k - \cosh k$
$\operatorname{arcosh}(3k) - k$
$-\operatorname{arcosh}(3k) - k$
x

Or


y
$3k - \frac{e^k + e^{-k}}{2}$
$\ln\left(3k + \sqrt{9k^2 - 1}\right) - k$
$-\ln\left(3k + \sqrt{9k^2 - 1}\right) - k$
x

12) c)



Or

